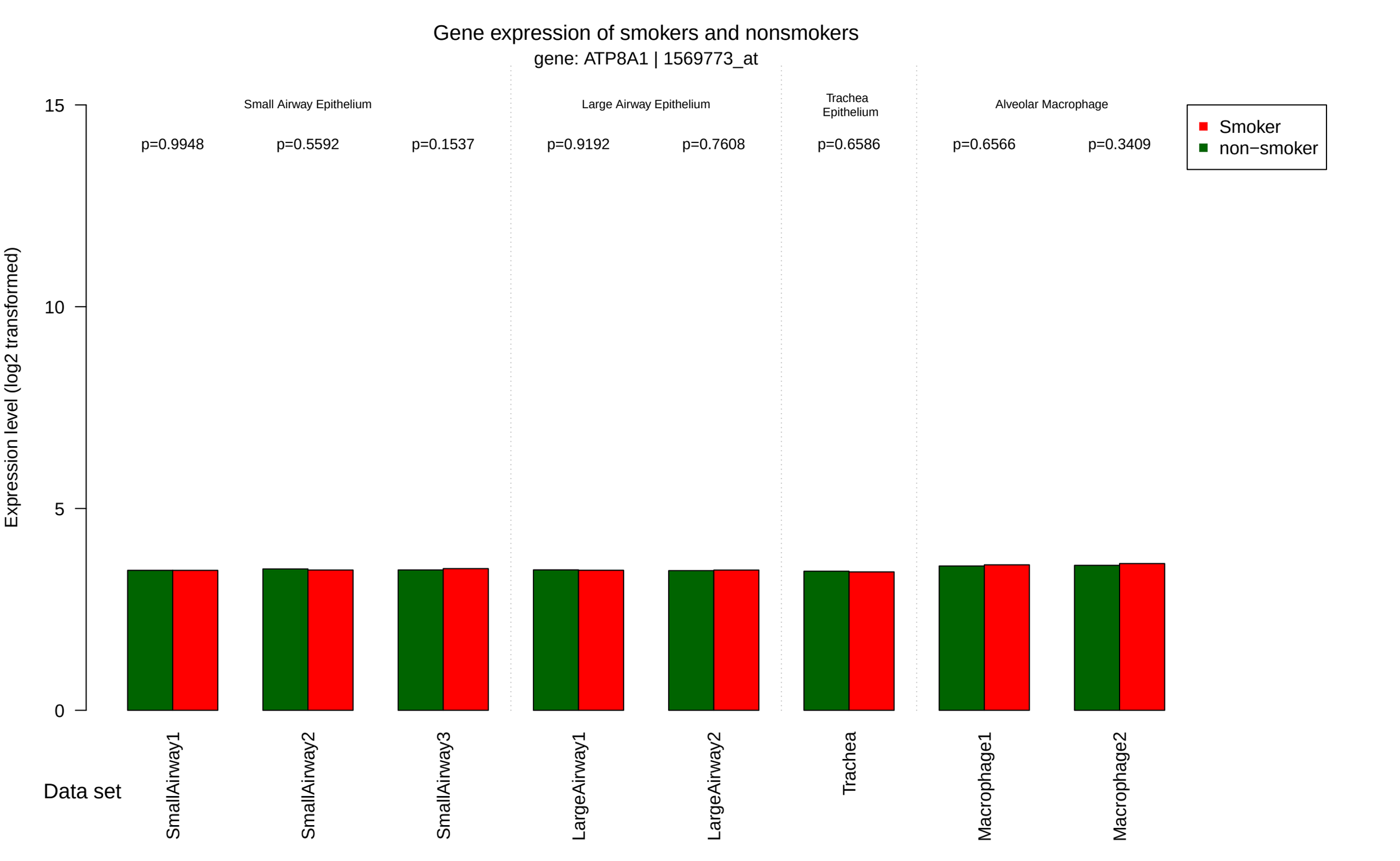
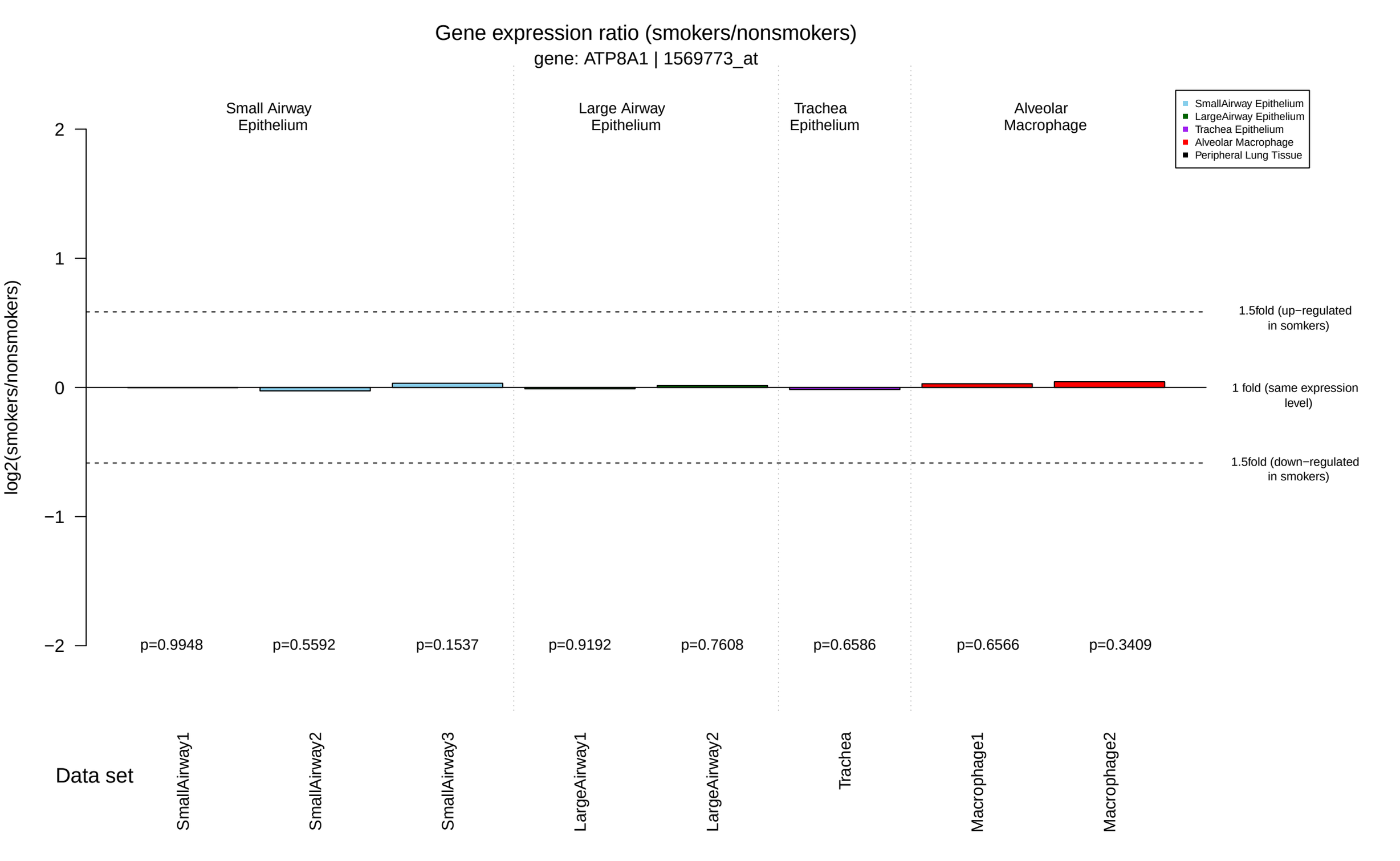
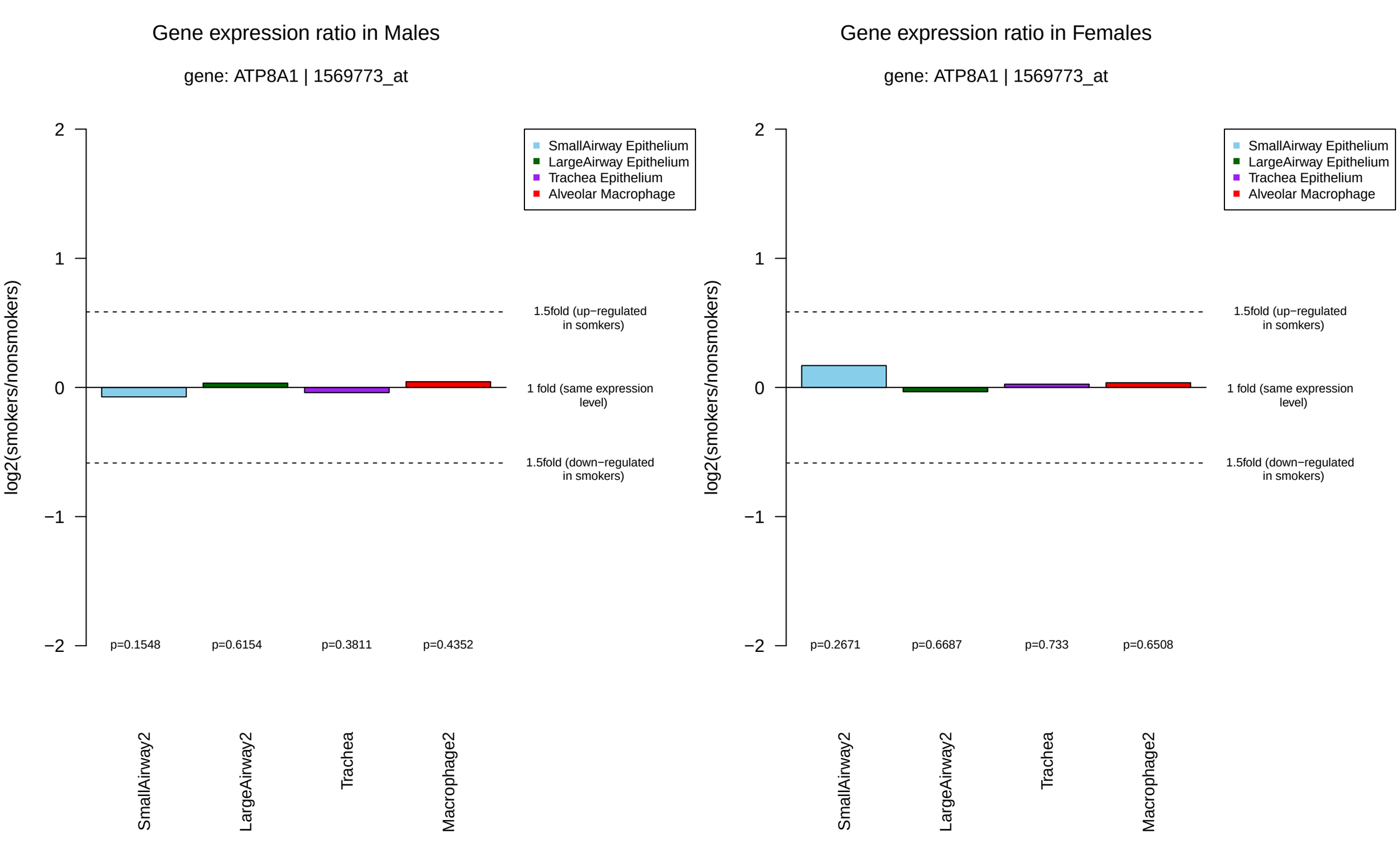
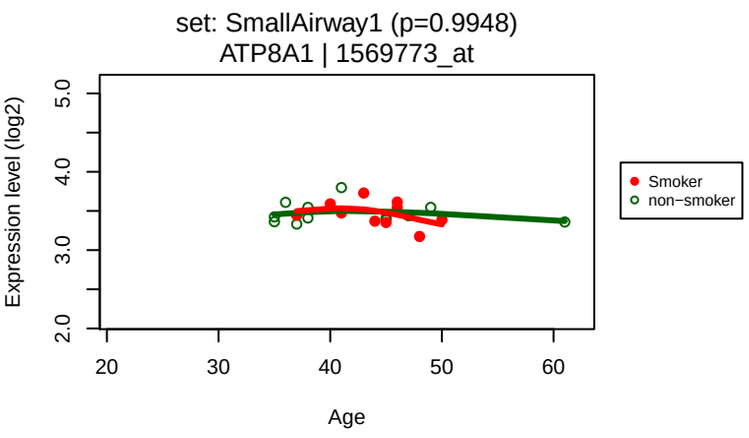
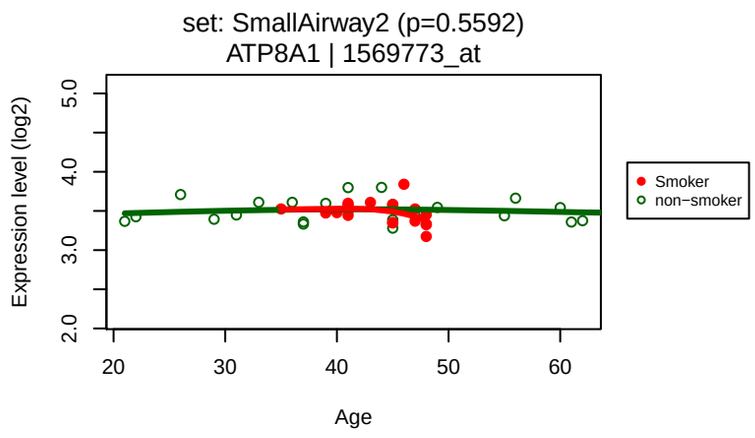
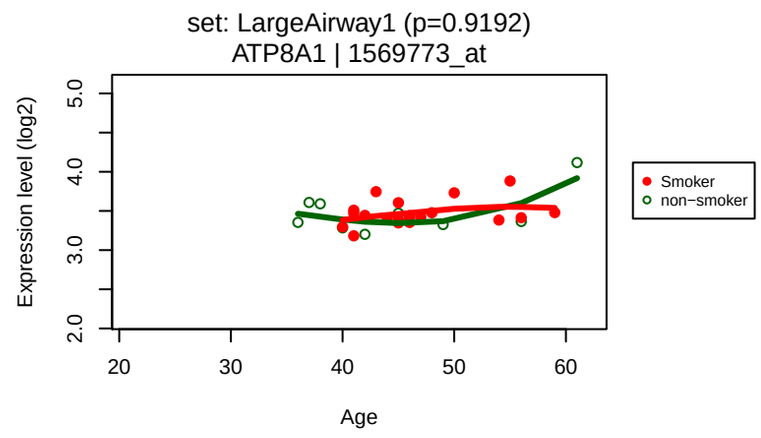

Gene expression of smokers and nonsmokers
gene: ATP8A1 | 1569773_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9948
p=0.5592
p=0.1537
p=0.9192
p=0.7608
p=0.6586
p=0.6566
p=0.3409
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

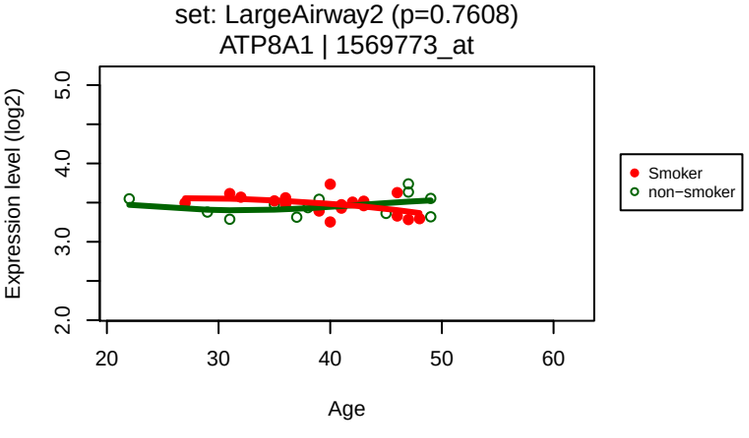

Gene expression ratio (smokers/nonsmokers)
gene: ATP8A1 | 1569773_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9948
p=0.5592
p=0.1537
p=0.9192
p=0.7608
p=0.6586
p=0.6566
p=0.3409
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

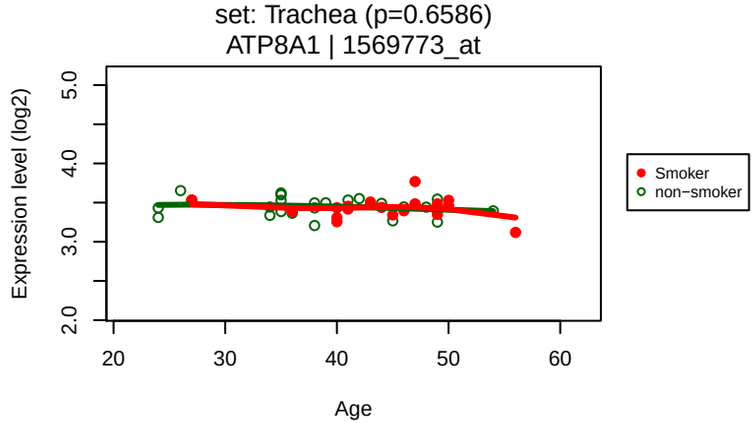

Gene expression ratio in Males
gene: ATP8A1 | 1569773_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1548
p=0.6154
p=0.3811
p=0.4352
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP8A1 | 1569773_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2671
p=0.6687
p=0.733
p=0.6508
SmallAirway2
LargeAirway2
Trachea
Macrophage2

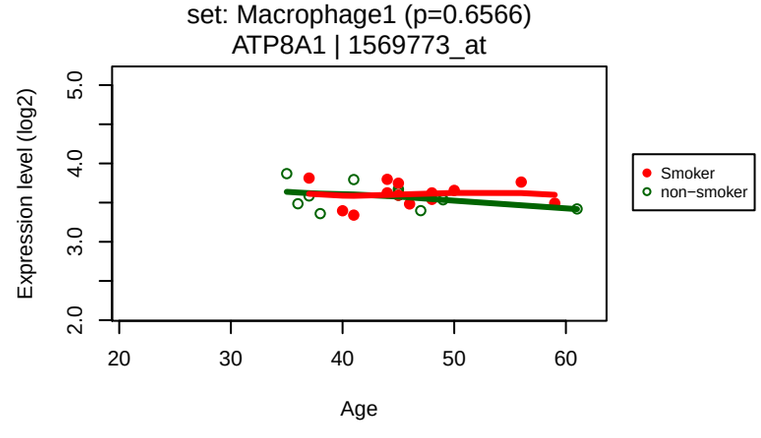

set: SmallAirway1 (p=0.9948)
ATP8A1 | 1569773_at
Expression level (log2)
Age
Smoker
non−smoker

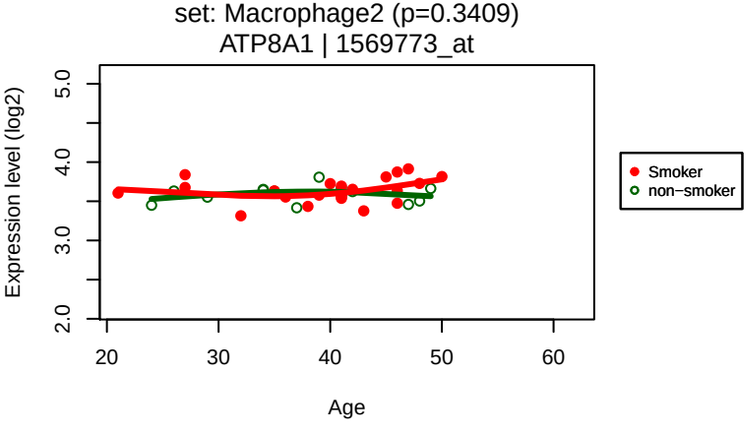

set: SmallAirway2 (p=0.5592)
ATP8A1 | 1569773_at
Expression level (log2)
Age
Smoker
non−smoker

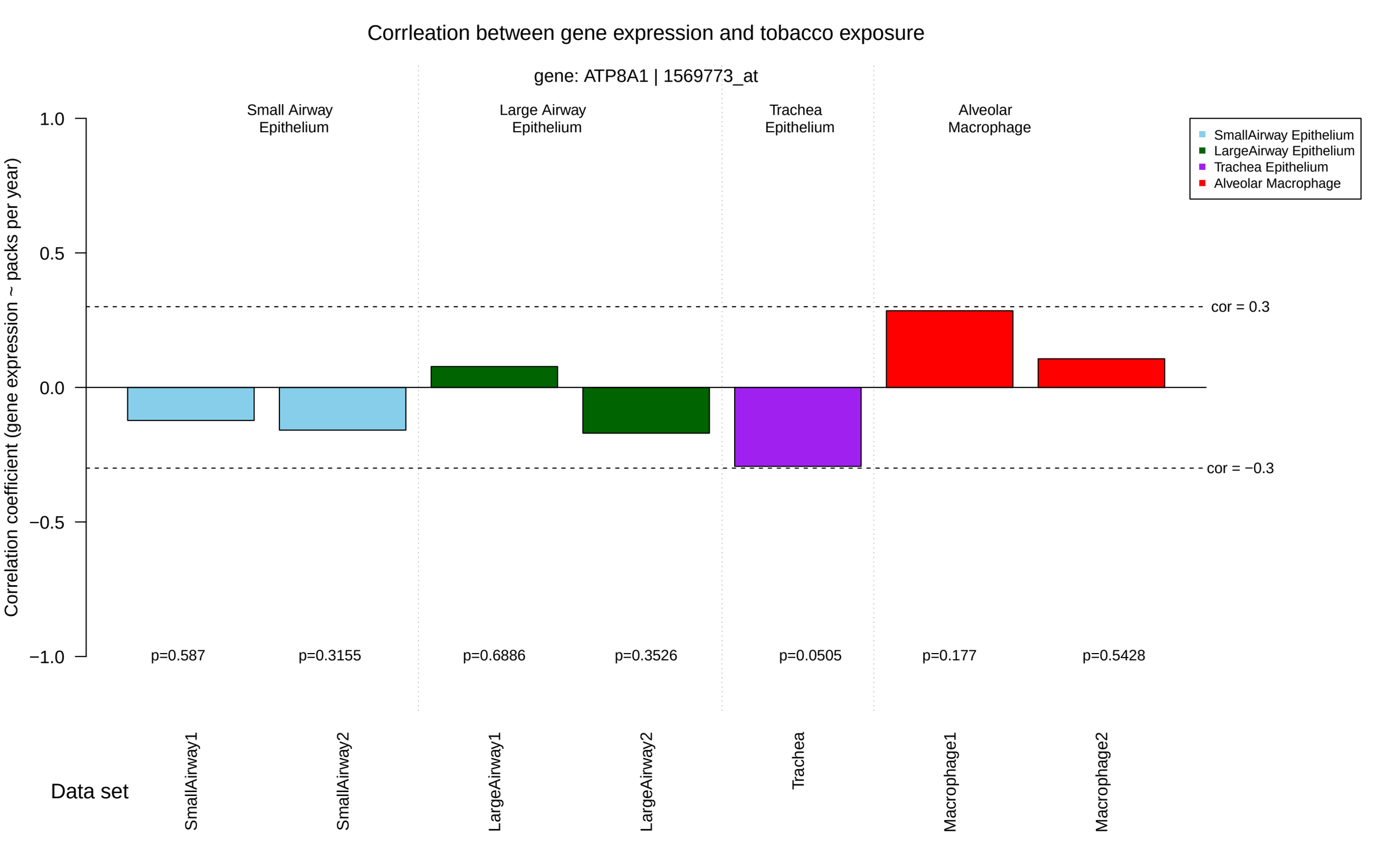

set: LargeAirway1 (p=0.9192)
ATP8A1 | 1569773_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7608)
ATP8A1 | 1569773_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6586)
ATP8A1 | 1569773_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6566)
ATP8A1 | 1569773_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3409)
ATP8A1 | 1569773_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP8A1 | 1569773_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.587
p=0.3155
p=0.6886
p=0.3526
p=0.0505
p=0.177
p=0.5428
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2